DISTRICT IQ

Technical Excellence Plan: Algorithms, Models and Architecture

Which technology to retain, fix, extend or replace, why, and how to prove it helps

Collector's District Intelligence Platform · Chennai District

Date: 8 October 2026 · Revision 2

Evidence scope: Audit report and Product Excellence Plan (8 Oct 2026); FarmwiseAI Task 6 brief; targeted reading of D:\farm (pipeline, search, assistant modules, label files); read-only queries on the live build of 8 Oct 2026 18:08 IST; primary external sources fetched 8 Oct 2026

Not done: No code, data or settings changed; nothing installed in the project; no model calls; no logged-in UI testing; no training or benchmark runs

Evidence labels: [V] verified · [S] static code · [D] document claim · [P] proposed · [H] hypothesis · [E] external source

Targets: All acceptance criteria are proposed targets, not measured results

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Evidence scope and how to read this document

What this is. This is an engineering specification of improvements to District IQ, the Collector's District Intelligence Platform for FarmwiseAI Task 6. It says which technologies, algorithms and models to keep, fix, extend or replace, why, and how to prove each change helps. It is a **recommendations document**. Nothing in it has been implemented. No code, data, settings or services were changed to write it, nothing was installed into the project, and no models were called.

Inputs actually used:

- The District IQ audit report and the Product Excellence Improvement Plan, both from 8 October 2026.
- The FarmwiseAI Task 6 brief: `Task_6 (2).pdf` and the challenge brief, pp. 20–23.
- Targeted reading of the repository at `D:\farm` on 8 October 2026, listed below.
- Read-only queries on the live data build cached on this PC (as of 2026-10-08 18:08 IST), and the pipeline's own evaluation report (`district_intel/output/reports/evaluation.md`, build of 4 October).
- Primary external sources, fetched on 8 October 2026 and listed in the References.

Code read for this plan (in addition to the audit):

- `district_intel/config.yaml`.
- Pipeline modules in `district_intel/dintel/`: `embed.py`, `eventmatch.py`, `placerole.py`, `newslocal.py`, `newsmodel.py`, `geolocate.py`, `geo.py`, `dedup.py`, `linking.py`, `newseval.py`, `severity.py`, `priority.py`, `analytics.py`.
- The label files in `district_intel/reference/labels/` and `district_intel/output/eval/`.
- Web app search and assistant modules: `src/lib/assistant/lance.ts`, `embed.ts`, `verify.ts`, `context.ts`, `types.ts`.

Most of these were read at the level of module documentation, function lists and key constants, not every line.

Not done: no live model calls, no logged-in walkthrough of the user interface, no load tests, and no training runs. Statements about model quality on District IQ data are therefore **hypotheses requiring evaluation** unless a measurement is cited.

Evidence labels used throughout:

Label	Meaning
[V] Verified	A query, test or request was run and its output seen (in the audit or for this plan)
[S] Static	Code or configuration was read but not executed
[D] Document claim	Stated in project documents, reports or notes; not independently re-checked
[P] Proposed	A design recommended here
[H] Hypothesis	Expected to help; needs a controlled evaluation
[E] External	From a primary external source (model card, documentation, paper), cited with access date

Audit finding IDs (for example `DATA-001`) refer to the audit report. New observations made for this plan are labelled `NEW-…`.

Synthetic data is permitted. The FarmwiseAI brief allows it for systems that could not be accessed (challenge brief, Task 6, p. 20: "Mock applications or sample exports may be used where live access cannot be provided."), and the team confirms FarmwiseAI told them to use it. This plan therefore never counts synthetic departmental data as a weakness. What it does require is (1) labelling on screen, so a demonstration record is never mistaken for a real event, and (2) not presenting accuracy measured against synthetic truth as real-world accuracy.

Revision 2 (8 October 2026) corrects the first version:

- **Synthetic data:** reframed as permitted.
- **Dates:** "today" now follows the current clock and discloses stale data.
- **Thresholds:** "measure, then set" targets replaced with provisional thresholds to agree before approval.
- **Sample sizes:** evaluation samples resized so each language subgroup is reported honestly.
- **Schedule:** the first-five and minimal-scope estimates now include the prerequisites they depend on.

1. Executive recommendation

1.1 The short answer

District IQ's architecture is better than its evidence. Three of its design choices are worth keeping:

- The AI never writes SQL or numbers.
- Tamil and English news reports of one event are matched by a staged pipeline (search, then reranker, then classifier, then an AI check on unclear pairs), with a rule that stops unrelated stories chaining together.
- Anomaly checks control false alarms statistically.

The weakness is that almost none of its accuracy has been demonstrated on **human-reviewed real data**, and the screens do not tell the Collector which records are synthetic stand-ins (permitted by FarmwiseAI) or how precisely each event is located.

So the first job is not new AI. It is honesty and measurement:

1. Show provenance and location precision everywhere.
2. Stop copying personal data to every computer.
3. Build small human-reviewed test sets from real Chennai records.
4. Retune the existing models against those sets.

Only then add new models, and only where a measured gap justifies them.

1.2 What this investigation found beyond the audit

These new observations change several earlier recommendations:

ID	Observation	Evidence
NEW-1	All 300 "hand labels" for the news incident filter were drafted by an AI (`labeled_by = claude-draft`) and none are human-reviewed (`reviewed_by` empty). The published news-filter score (F1 0.63) therefore measures agreement with AI drafts, not accuracy.	[V] label file read
NEW-2	The real-data cross-source link label file has 141 sampled pairs and **0** answers in `same_incident`. The test-set gold columns (`news_test_set.csv`, 199 rows) are also empty.	[V]
NEW-3	The project's notes say the place and same-event label sets (320 + 120 places, 305 pairs) were also labelled by Claude. The pair file has no labeller column. Reported gains (places 69% → 89% correct; Tamil–English grouping recall 1% → 80% at 92.5% precision) are cross-validated against those AI-drafted labels.	[D] notes; [V] no labeller column
NEW-4	In the 4 October build, 6,984 of 9,953 news documents (70%) had no confident label (`review`), because the local classifier was unsure and the free-tier LLM had not reached them. Incident articles may therefore be undercounted: notes record incident events dropping from 858 to 585 in one build.	[V] evaluation report; [D] notes
NEW-5	The pipeline's high scores are measured against synthetic truth: within-source dedup B-cubed F1 0.996 (generator's own truth); cross-source linking F1 0.861 (held-out planted events); category macro-F1 0.995 (templated synthetic complaints). They say nothing about real data.	[V] evaluation report
NEW-6	Three different embedding models are in use: multilingual-e5-small (pipeline linking), multilingual-e5-base (news classifier option, web search), bge-m3 (event grouping).	[S]
NEW-7	The web search index stores no provenance field (real or synthetic) and news rows carry no zone (`zone_no = -1`), so search cannot filter by real-only or by area for news.	[S] `lance.ts`
NEW-8	The number verifier's `Fact` type holds only id, label, value and unit: no filters, time window, unit of counting or denominator.	[S] `types.ts`
NEW-9	Priority weight learning has 0 Collector decisions to learn from (it needs 30).	[V] evaluation report
NEW-10	Geographic resolution from text was measured only on synthetic grievances (ward accuracy 0.474, median error 1,028 m), whose street names the README says are not tied to their map pins. The figure is not evidence for real text.	[V] report; [D] README
NEW-11	The AWS-mode store reload is recorded in project notes as blocking the web server 4–7 s every 5 minutes (30–40 s on the hourly build).	[D] notes, not re-measured

1.3 Corrections to the earlier improvement plan

Earlier recommendation	Problem	Corrected recommendation
Merge near-identical headlines within 72 h (#11)	Identical headlines are not proof of one event. "Woman killed as lorry hits two-wheeler in Chennai" can describe different accidents on the same day. The existing grouping already uses place, time, deaths and a reranker, plus anti-chaining.	Merge automatically **only exact copies** (same canonical URL, or the same publisher with identical body). Send cross-outlet identical headlines through the existing event matcher with place and time agreement. First find out *why* 47 duplicate titles remain (unprocessed pair backlog vs genuinely different events).
Split API keys to protect personal data (#2)	Separate keys still copy citizens' data to each computer if the data is in the download. The website's login also needs the `users` table locally.	Move personal data and login behind one central server. Teammates' copies receive only de-identified intelligence tables and authenticate against the central service (section 6).
Measure accuracy on labelled sets (#10)	It did not notice that the existing labels are AI drafts (NEW-1, NEW-3).	Human review is a prerequisite: two people label a stratified sample, and disagreements are adjudicated. AI drafts may only pre-fill.
"Serious" lane trigger: two or more outlets (#5)	Syndicated copies (one agency story republished) are not independent corroboration.	Count **independent outlets** (distinct publisher groups, after removing wire copies and identical bodies). Show "1 report, republished by 3 outlets" honestly.
"In the news, missing from departments" board (#16)	A news story with no matched record does not prove the department has no record: departmental data is synthetic, and linking recall on real data is unmeasured.	Label it "**no matching record found**", and show linking recall next to it once measured.
Rain-to-complaint early warning (differentiator)	Rain factors are learned from synthetic complaint history.	**Defer** until a real complaint feed exists. Meanwhile, show only real rainfall and river-level alerts.
Intervention check (differentiator)	Before/after change is not causal.	**Defer**; specify a difference-in-differences design with an "insufficient evidence" outcome for when real data exists (section 4.I).

1.4 What to do, grouped

A. Immediate correctness and security fixes (no new models)

- Provenance end to end (`DATA-001`): screens, exports, search index and assistant facts.
- Precision-aware geography: no exact pins for unknown places; an "unlocated" bucket (`GIS-001`).
- Central personal-data handling and authentication (`SEC-001`); a private, authenticated upload (`SEC-002`); one safe outbound fetch (`SEC-003`).
- One effective-status rule for every count (`KPI-001`); no writes on page views (`BE-001`); durable audit (`OPS-001`).
- Fix guard false refusals (`AI-001`) and the failing pipeline tests (`TEST-001`).

B. Algorithm and intelligence improvements (mostly *better data*, then retuning)

- Human-reviewed gold sets for incident filter, category, event place, same-event pairs, cross-source links, retrieval and assistant answers.
- Retrain the existing local news classifier on reviewed labels to shrink the 70% "review" backlog (NEW-4).
- Retune event grouping and linking thresholds on reviewed pairs.
- Typed, scoped facts and claim-level verification for the assistant (NEW-8).
- Typed conversation state with explicit reset.

C. Operational workflow improvements

- A "serious, not yet confirmed" lane with independence-aware corroboration.
- "What changed since the last briefing".
- Assignment deadlines, escalation and closure evidence.
- Per-item freshness.
- Unified search that respects permissions and provenance.

D. Advanced differentiation (after A and B)

- An evidence-graded briefing.

- A Tamil voice brief.
- Intervention analysis and rain early warning: designed now, switched on only with real departmental data.

1.5 What needs better data rather than better AI

Problem	Why AI won't fix it	What fixes it
Severe and multi-source incidents are synthetic (permitted)	No model can turn demonstration records into real events	Label them on screen now; connect real departmental feeds when available
Linking accuracy unknown	A bigger model cannot be validated without real truth	150–300 human-labelled real pairs
Taluk approximate	Boundaries are a reference-data problem	Official taluk polygons
Priority learning idle	Needs Collector decisions as outcomes	Real use; keep explainable rules meanwhile
Early warning	Trained on synthetic history	Real complaint history

1.6 What must be proven before real Collector use

Proposed gates; targets, not results:

1. Every incident-bearing view shows provenance, and a real-only mode exists.
2. No personal data or password hashes exist outside the central server.
3. On human-reviewed real samples, report with confidence intervals:
    - incident-filter precision and recall;
    - event-place accuracy;
    - same-event grouping precision and recall;
    - cross-source link precision and recall.
4. On a frozen dataset, the assistant answers a reviewed question set with correct numbers, correct scope and supported citations, and refuses safely.
5. A tile, a list, the assistant and an export give the same count for the same scope.

2. Current technology stack

2.1 Stack inventory

Layer	Current technology	Purpose	Implementation evidence	Limitation	Recommendation
Frontend	Next.js 14, React 18, TypeScript, Tailwind	Citizen, officer and Collector portals	[S] `package.json`; `src/app/*`	One large console data module (`intel.ts`, 1,432 lines)	Retain; split read models by page
Charts	ECharts 6 (`echarts-for-react`)	KPI and trend charts, assistant charts	[S] deps; `ChartRenderer.tsx`	—	Retain
Maps	Leaflet 1.9	Ward and zone map, pins	[S] deps; `SatMap.tsx`	Draws centroid fallback as exact pins (`GIS-001`)	Retain; add precision rendering
Backend/API	Next.js route handlers (Node 22), Zod validation	71 API routes	[S] audit route inventory	GET with writes (`BE-001`); uneven validation	Retain; read-only GETs
Python pipeline	pandas, numpy, scipy, scikit-learn, sentence-transformers, torch (CPU)	Build the curated store	[S] `dintel/*`, `requirements.txt`	Model pickles saved with older scikit-learn (`DATA-003`)	Retain; pin versions
Database (pipeline)	SQLite (`district_intel.db`, 33 tables)	Curated build	[V]	—	Retain
Database (web)	Two modes: MySQL 8, or `DATA_BACKEND=aws` = in-memory SQLite plus a MySQL→SQLite dialect translator	Serve the build	[S] `lib/aws/store.ts`, `dialect.ts`; [V] `.env` sets aws	Dialect drift (`CODE-001`); READ ONLY is a no-op (`AI-003`); reload pauses (NEW-11)	Pick one serving engine (section 6)
Durable store	DynamoDB via team API, mirrored to local `durable.sqlite`; one saving PC	Users, complaints, decisions, chats	[S] `store.ts`; [V] durable file columns	PII and hashes on every copy (`SEC-001`); single writer	Central server
AWS services	S3 snapshot, API Gateway + Lambda collectors, DynamoDB	Hourly refresh and storage	[S] `.github/workflows/aws-refresh.yml`	One shared key	Per-service credentials
Embeddings (pipeline)	`intfloat/multilingual-e5-small` (384-d) for linking and classifier features; `BAAI/bge-m3` (1024-d) for same-event candidates	Meaning across Tamil and English	[S] `embed.py`, `eventmatch.py`	Three models in use (NEW-6)	Evaluate consolidation
Embeddings (web)	`Xenova/multilingual-e5-base` (768-d) via transformers.js / ONNX in a worker	Assistant semantic search	[S] `assistant/embed.ts`, `lance.ts`	Separate vectors from the pipeline	Evaluate against bge-m3 on real queries
Vector + keyword search	LanceDB (embedded): vector cosine + BM25 full-text (fuzzy), rank fusion	Assistant retrieval	[S] `lance.ts`; [E] LanceDB docs (RRF default)	No provenance field; news rows unlocated (NEW-7)	Extend index fields
Reranker	`BAAI/bge-reranker-v2-m3` cross-encoder (pipeline only)	Score same-event pairs	[S] `eventmatch.py`; [E] model card	CPU throughput ~6 pairs/s [D]	Retain
LLMs (assistant)	Groq: openai/gpt-oss-20b, openai/gpt-oss-120b, qwen/qwen3.8-27b; Gemini flash family fallback; Bedrock option	Router, planner, composer	[S] `lib/ai/gateway.ts` defaults; `.env` AI_PROVIDER=groq, fallback gemini	Free-tier limits; [E] Groq page lists 8K tokens/min base	Retain the pool; measure schema validity
LLMs (pipeline)	Groq gpt-oss-120b → 20b for news classify and explain; Gemini flash chain for briefs	Labels, details, narratives, translation	[S] `config.yaml news_llm`	Quota leaves 70% unlabelled (NEW-4)	Use LLM as teacher and grey-band judge only

Layer	Current technology	Purpose	Implementation evidence	Limitation	Recommendation
Classifiers	TF-IDF (word + char n-gram) + logistic regression cascade; optional SetFit on e5-base; char n-gram category classifier	News relevance and category; complaint categories	[S] `newslocal.py`, `newsmodel.py`, `classify.py`	Trained on LLM labels and synthetic text	Retrain on reviewed gold
GIS	Point-in-polygon on 200 GCC wards (numpy + matplotlib.path), snapping ≤500 m, gazetteer with Tamil aliases, place-role model	Locate events	[S] `geo.py`, `geolocate.py`, `placerole.py`	Taluk by majority vote (`GIS-002`); centroid fallback	Retain; official taluks
Statistics	Poisson baseline (weekday + rain), CUSUM, Benjamini–Hochberg FDR, DBSCAN, Getis–Ord Gi*, EWMA	Anomalies and hotspots	[S] `analytics.py`	Baselines from synthetic counts	Retain; add coverage guard
Scheduling	GitHub Actions cron → AWS Lambdas; PC Task Scheduler (`DistrictIntel`, noted as disabled [D]); background sweeps in web requests	Refresh	[S] workflow; `runDueSources()`	Refresh triggered inside requests	Move to a job runner
Auth	JWT (jose, HS256) in per-portal cookies; bcrypt; role checks per route	Sessions and roles	[S] `jwt.ts`, `auth.ts`, guards	Fallback secret (`SEC-004`); local users table	Central auth
Audit	`audit_log` table, best-effort insert	History of actions	[S] `sources.ts audit()`	Silent failure (`OPS-001`)	Transactional audit
Tests / eval	Vitest (330 pass), pytest (88/90 and 90/90), tsc; pipeline `evaluation.md`; `news-eval`	Regression and accuracy	[V] audit	Gold labels AI-drafted or empty (NEW-1, NEW-2)	Human-reviewed eval harness

Present in dependencies, not confirmed as actively used:

- `tesseract.js` (OCR; an OCR source route exists).
- `@aws-sdk/client-bedrock-runtime` (configured as an option; notes say the keys expired [D]).
- `@ai-sdk/openai` (fallback option).
- `playwright-core` (Studio browser connector [S]).

None of these was exercised in this investigation.

3. Retain, improve or replace matrix

Subsystem	Existing approach	Decision	Evidence	Preferred improvement	Why	Effort	Risk	Validation
Assistant query safety	LLM fills a plan; compiler emits parameterised SELECT over a catalog	**Retain**	[S] `compile.ts`	Add read-only connection plus cancellation in AWS mode	Sound design; enforcement gap only	M	Low	Write attempt fails; heavy query cancelled
Number verifier	Value match with measure words and direction; `{{fact_id}}` citations	**Extend**	[S] `verify.ts`, `types.ts`	Scoped facts plus claim-level checks (section 4.F)	A matching value can still be the wrong scope	M	Medium	Reviewed answer set: scope errors = 0
Code guard	Regex refusals before any model	**Fix implementation**	[V] 2 false refusals	Command-form rules; data-word exemptions; tests	Cheap, effective, brittle	S	Medium	0 false refusals on near-miss set
Intent fallback	Keyword intents when LLM unavailable	**Fix / extend**	[V] misroutes brief examples	Add brief questions and chains to eval; refinement rules	Fallback is used often on free tier	S–M	Medium	Routing accuracy on reviewed set
News relevance and category	TF-IDF+LR cascade → Groq LLM; SetFit option	**Extend** (retrain on gold); **evaluate alternative** (MuRIL)	[V] 70% in review; [S]	Retrain local heads on reviewed labels; LLM only for grey band	Coverage bottleneck is labels and quota, not model size	M	Medium	Precision and recall on reviewed test set; review share
Entity, deaths, status extraction	LLM details on headlines/snippets	**Fix implementation**	[V] 98% snippets; [D] animal deaths counted (notes)	Schema output with evidence spans; reject unsupported fields	Headlines rarely support status or counts	M	Low	Field accuracy on reviewed set
Event place	Role cues + LR, district fallback	**Retain**, re-validate	[S] `placerole.py`; [D] CV vs AI labels	Human-review labels; show role and precision	Good fit for Tamil and English cues	S	Low	Place accuracy, wrong-place rate
Same-event grouping	bge-m3 → reranker → sign-constrained LR → LLM grey band → constrained agglomeration	**Retain**; retune	[S] `eventmatch.py`; [V] 2 failing tests	Exact-copy dedup first; clear the pending-pair backlog; thresholds from reviewed pairs	Architecture is appropriate; evidence is not	S–M	Medium	Pair and cluster precision/recall on reviewed set
Within-source dedup	Union-find with blocking (synthetic)	**Defer**	[V] scored vs generator truth only	Re-evaluate when real feeds arrive	Nothing real to dedupe yet	—	—	—
Cross-source linking	Blocking + LR trained on planted pairs	**Fix** (retrain on real labels)	[V] 0 real labels	Label 150–300 real pairs; recalibrate bands	Synthetic training cannot transfer	M	Medium	Real-pair precision/recall
Geo: ward/zone	Point-in-polygon, snapping	**Retain**	[V] 0 ward/zone mismatches	Store uncertainty radius; no pin when coarse	Correct method	S	Low	Pin precision rule
Geo: taluk	Majority vote of synthetic coded points	**Replace, with evidence**	[V] Kolathur 1 ward; [S]	Official taluk polygons, point-in-polygon	Reference-data error	M	Medium	100% assignment by polygon
Search	e5-base vectors + BM25 + RRF (LanceDB)	**Extend**; **evaluate alternative** embedding	[S] `lance.ts`	Provenance, geography and permission fields; exact-name boost; evaluate bge-m3	Gaps are fields and filters first	M	Low	Top-5 hit rate on reviewed queries
Severity	Static rules from facts	**Retain**; separate axes	[S] `severity.py`	Severity vs urgency vs confidence (section 4.G)	Explainable; mixing axes misleads	S–M	Low	Reviewer agreement
Priority	Hand-set weights; MAP learning after 30 decisions	**Retain rules**; **defer learning**	[V] 0 decisions	Collect decisions with reasons	No outcome labels yet	—	—	AUC once ≥ 30 decisions
Anomalies	Poisson + CUSUM + BH FDR	**Retain**; extend	[S] `analytics.py`	Coverage-change guard; real-only series	Statistically sound	S–M	Low	Alert precision reviewed weekly

Subsystem	Existing approach	Decision	Evidence	Preferred improvement	Why	Effort	Risk	Validation
Hotspots	DBSCAN + Gi*	**Retain**; precision filter	[S]	Exclude coarse points	Sound	S	Low	No hotspot from centroid points
Store serving (AWS mode)	In-memory SQLite + MySQL dialect translation	**Evaluate alternative**	[S]; [D] reload stalls	One engine (section 6)	Two dialects double the risk	L	High	Parity tests; reload latency
Durable writes	Single saving PC	**Replace**	[S] README	Central API service	Data loss and PII risk	L	High	Concurrency tests
Overlay (scenario)	Planted synthetic cross-source events	**Retain for testing only**; off for demos and evaluations	[S] `world.py`	Separate "scenario" build	Prevents mistaking synthetic for evidence	S	Low	Real-only build has 0 overlay rows

4. Detailed algorithm recommendations

Each card states the administrative question, the current state with evidence, the recommendation and how to prove it. Acceptance criteria are **proposed targets**. Numeric thresholds refer to the provisional table in section 7.1 (T1–T12), which should be agreed before work is approved.

4.A News relevance and incident extraction

Field	Detail
Administrative question	"Which of today's Chennai news reports describe an event a department must act on, what is it, where and when did it happen, how serious is it, and is it ongoing?"
Current implementation	[S] Cascade in `newslocal.py`: word + character TF-IDF with logistic regression heads (in Chennai? incident? report type? category?). Low-confidence items go to the Groq LLM (gpt-oss-120b, then 20b per `config.yaml`), whose cached answers become training labels. An optional SetFit model on multilingual-e5-base exists (`newsmodel.py`). Details (people, organisations, deaths, injured, status, place) come from an LLM.
Measured weakness / uncertainty	[V] 70% of documents were unlabelled ("review") in the 4 Oct build. [V] SetFit agreement with the LLM: category 0.756, report type 0.62 (agreement, not accuracy). [V] All 300 incident labels are unreviewed AI drafts. [V] 98% of news rows are Google News snippets; only 790 have full text. [D] Notes record animal deaths counted as deaths.
Recommended approach	**Separate the questions and keep the cascade:** (1) district relevance; (2) report type (incident, civic complaint, announcement, court, statistics…); (3) category; (4) entities and counts; (5) event place (4.B); (6) event time; (7) status. Steps 1–3 stay with the local classifier, **retrained on human-reviewed gold labels** and calibrated. Steps 4, 6 and 7 use schema-constrained LLM extraction **only on full text or a reviewed snippet**, with an evidence span (exact quoted text) required for every field; a field without a span is stored as unknown.
Why it fits	Cost and quota are the binding constraints (free tier, CPU only). A calibrated local classifier handles volume. The LLM handles ambiguity. Evidence spans make extraction checkable.
Alternative and trade-offs	Fine-tune MuRIL [E] (Apache-2.0, includes Tamil and transliterated text) as the local classifier. It may handle Tamil and Tanglish better than TF-IDF [H], but needs GPU-hours or slow CPU training and more labels. **Benchmark first** on the same gold set.
Inputs and labels	600 reviewed articles stratified by language (Tamil, English, Tanglish), publisher tier, and classifier confidence band. Per article: relevance, report type, category, and a death/injury count with span.
Processing steps	Canonicalise URL → language detect → relevance head → report-type head → category head → if incident and full text available: LLM extraction with spans → validators (numbers in span; animal vs human words; Tamil count words) → store fields with method, model and confidence.
Output and evidence	Each field with value, method (rule, classifier, LLM), model id, confidence, evidence span, and source document id.
Uncertainty and missing data	Snippet-only articles: relevance and category only; deaths and status left unknown unless stated in the snippet. Confidence under threshold → review queue, not silent "OTHER".
Event time vs publication time	Store `published_at` (from the feed), `collected_at` (fetch time), and `event_time` with a precision: exact, day, "yesterday" relative to the publication date, or unknown. Never copy publication time into event time silently. Incidents should sort by event time when known, otherwise by first publication, and say which.
Failure modes	LLM quota exhausted (fall back to local, mark "unconfirmed"); syndicated duplicate bodies; Tamil numerals in words; headlines about other districts with a Chennai dateline.
Existing modules	`district_intel/dintel/newslocal.py`, `newsmodel.py`, `newsllm.py`, `loaders/news.py` [S]
Migration	Keep the current labels as "silver". Add a reviewed "gold" set, kept out of training for the test portion. Retrain and compare on the gold test set before switching.
Evaluation	Precision, recall and F1 for "Chennai incident" on the gold test set; macro-F1 for category; field-level accuracy for extraction; the "review" share.

Field	Detail
Proposed acceptance	Provisional (T1–T3, section 7.1): Chennai-incident precision ≥ 0.85 and recall ≥ 0.75 on the gold test set; category macro-F1 ≥ 0.70; review share ≤ 30% with precision not lower than the current cascade beyond its interval. Every metric reported with sample size and 95% interval.
Effort, dependencies, risk	M (3–5 dev days plus labelling). Depends on gold labels (section 7). Risk: retraining can shift counts; compare before and after on a frozen build.
Decision	**Implement now** (labels and retraining); **benchmark first** (MuRIL).

4.B Geographic resolution

Field	Detail
Administrative question	"Where exactly did this happen: which ward, zone and taluk, and how sure are we?"
Current implementation	[S] Gazetteer with Tamil aliases → place-role model (`placerole.py`: event, hospital, title, origin, style, route, list, mention; logistic regression; below a minimum probability the report stays at district level) → point-in-polygon on 200 GCC wards (`geo.py`), snapping within 500 m → ward→taluk by majority vote of coded synthetic points (`geolocate.py`).
Measured weakness / uncertainty	[V] 501 of 856 real news incidents at the city centroid with 15 km precision, drawn as pins. [V] Taluk derived from synthetic data. [D] Place accuracy 89% under cross-validation against AI-drafted labels. [V] Text-only ward accuracy measured only on synthetic grievances.
Recommended approach	**Retain the role model and point-in-polygon. Add explicit uncertainty and keep geographies distinct.** Store `geo_level` (point, street, locality, ward, zone, district, outside, unknown) and an uncertainty radius for every record. Derive ward, zone and taluk only when the uncertainty fits inside the polygon; otherwise leave them null and count the record as "unlocated". Use official taluk polygons for taluk, independent of GCC zones and wards.
Why it fits	The failure is presentation and reference data, not the matching algorithm.
Alternative and trade-offs	An external geocoder (for example a hosted geocoding API) for street-level text. It may raise coverage [H], but Chennai locality names are ambiguous, licences vary, and it sends text outside. **Defer** until the gazetteer's misses are measured.
Inputs and labels	300 real news reports reviewed for event place (or NONE, or outside the district), with role; 100 held out from a later week.
Processing steps	Extract mentions → assign roles → choose event place if probability ≥ threshold → resolve to a polygon or point with a radius → set the administrative keys only if contained → flag out-of-district (and exclude from district totals).
Output and evidence	Event place, its role, all other places with roles, method, radius and confidence.
Ambiguity	Names shared across districts (e.g. a locality name existing in Chennai and Chengalpattu) → "ambiguous", review queue. District-wide events (rain warnings) → `geo_level = district`, no pin.
Failure modes	Dateline mistaken for event place; hospital as event place; transliteration variants; GCC zone mistaken for revenue taluk.
Existing modules	`placerole.py`, `geo.py`, `geolocate.py`, `loaders/news.py`; web `intel.ts` pin query, `SatMap.tsx` [S]
Migration	Add the columns; backfill from the existing `loc_precision_m` and `geo_level`; change map rendering. No change to matching.
Evaluation	Event-place accuracy, wrong-place rate (worse than leaving it unknown), coverage (share located to ward), and median error where a gold point exists.
Proposed acceptance	0 pins with uncertainty above about 2 km; wrong-place rate ≤ 10% (T4) with its interval; 100% of out-of-district records excluded from district totals.
Effort, dependencies, risk	S for uncertainty and display; M for official taluks (needs boundary file).
Decision	**Implement now** (uncertainty, display, labels); official taluks when the file is available.

4.C Duplicate detection and incident clustering

Five different relationships must be kept apart; they are not one "duplicate" problem:

Relationship	Example	Correct handling
Copy of the same article	The same URL via Google News and the publisher	Merge automatically (canonical URL, or same publisher with identical body)
Different reports of one event	The Hindu and Dinamalar on the same fire	Group into one story/incident; keep each report as evidence
Update to the same event	A later report raising the death toll	Same incident; add a timeline step; keep the earlier figure
Related but distinct events	Two accidents on the same road on the same day	Separate incidents; may share a "related" link
Recurring problem	Waterlogging in the same street every rain	Separate incidents; a recurrence link and count over 90 days

Field	Detail
Administrative question	"How many distinct things actually happened, and which reports belong to each?"
Current implementation	[S] Exact and near-copy handling in the news loader. `eventmatch.py`: bge-m3 neighbours within a time window (cosine ≥ 0.55) → bge-reranker-v2-m3 cross-encoder → 13 features (cosine, margin, reranker score, place agreement/conflict, category, time gap, death-count change, shared numbers) → sign-constrained logistic regression → merge if p ≥ 0.80, LLM check if 0.35–0.80, separate otherwise → constrained agglomeration (mean between-group p ≥ 0.50, no pair below 0.08, max group size). Incident IDs are kept stable across builds (`linking.stable_ids`).
Measured weakness / uncertainty	[V] 47 identical titles across different news incidents in the live build. [V] 2 failing tests, one showing an extra Tamil–English merge. [D] About 12k low-priority and 8.3k unsure pairs not yet reranked or LLM-checked. [D] Precision and recall measured only against AI-drafted pair labels.
Recommended approach	**Retain the architecture.** (1) Automatic merge only for exact copies. (2) Clear the pending-pair backlog before judging the 47 duplicates. Classify each into backlog, genuinely different event, or algorithm miss. (3) Recalibrate thresholds on human-reviewed pairs with hard negatives (same place and day, different event). (4) Add an **update** link type: same event with a later timestamp and changed counts. (5) Split/merge review in the UI. A merge keeps both old IDs as aliases pointing to the surviving ID; a split issues a new ID and records its parent.
Why it fits	The staged design is the established pattern for this problem (cheap retrieval, expensive verification, anti-chaining), and it already handles Tamil–English pairs.
Alternative and trade-offs	Plain HDBSCAN or connected components on embeddings: simpler, but prone to chaining unrelated events. **Not recommended.**
Inputs and labels	400 reviewed pairs: 25% Tamil–English, 30% hard negatives, sampled from different cosine bands; a cluster-level sample of 60 stories reviewed whole.
Output and evidence	Story ID with member reports; per pair the probability, features and LLM verdict; an audit of merges and splits.
Uncertainty	The grey band stays separate until confirmed (no LLM answer → separate). Show "possibly the same as…" links instead of merging.
Failure modes	Syndicated copies counted as corroboration; anniversary or follow-up court stories merged with the original event; chaining through a vague report.
Existing modules	`district_intel/dintel/eventmatch.py`, `loaders/news.py`, `linking.py` (stable IDs) [S]
Migration	Keep current story IDs; alias table for any re-grouping; re-run on a frozen build and diff.
Evaluation	Pairwise precision and recall; cluster-level B-cubed precision and recall; false-merge rate on hard negatives; Tamil–English recall separately.
Proposed acceptance	False-merge rate on hard negatives ≤ 5% (T5), with interval; no unexplained identical-title duplicates after backlog processing; IDs stable across 3 rebuilds of an unchanged dataset.
Effort, dependencies, risk	S–M. Depends on reviewed pairs and the failing-test fix. Risk: threshold changes shift story counts; diff before switching.

Field	Detail
Decision	**Implement now** (exact-copy rule, backlog, review UI); **benchmark first** (threshold changes).

4.D Search and retrieval

Field	Detail
Administrative question	"Find everything we have about Pattinapakkam / this outlet / this department, quickly, without showing what I'm not allowed to see."
Current implementation	[S] LanceDB embedded index with two collections (incidents, news). Vectors from multilingual-e5-base (768-d) in a worker thread; BM25 full-text with fuzziness 1; reciprocal-rank fusion (RRF); filters on time, zone, category, severity and department. Counts always come from SQL, not retrieval.
Measured weakness / uncertainty	[S] No provenance field; news rows have no zone; no added-source items or Studio datasets; [R2/R11] search PARTIAL. No retrieval evaluation set exists, so retrieval quality is **unknown**.
Recommended approach	Fix the fields before the model. (1) Add `provenance`, `geo_level`, `ward`, `zone`, `taluk`, `published_at`, `collected_at` and `visible_to` (role or department) to every record. (2) Apply permission and provenance as **pre-filters** inside the query, not after ranking. (3) Add an exact-match lane for IDs, place names and outlet names before fusion. (4) Collapse results by story ID so syndicated copies don't fill the list. (5) Index news documents, source items and Studio rows.
Why it fits	The demonstrated gaps are missing fields and filters, not embedding quality.
Alternative and trade-offs	Replace e5-base with bge-m3 (MIT, 1024-d, dense, sparse and multi-vector, 8,192 tokens [E]) so web search and pipeline grouping share one model. Possible quality gain [H]; costs: larger vectors, slower CPU encoding (notes: about 4.5 texts/s for bge-m3 on this PC [D]), and a re-index. Optional cross-encoder reranking of the top 20 with bge-reranker-v2-m3 (LanceDB supports cross-encoder rerankers [E]). **Benchmark first.**
Inputs and labels	60 real queries (20 English, 20 Tamil, 20 Tanglish) with judged relevant items per query.
Index refresh	Keep the existing hash-based sync. A record removed from the store is deleted from the index in the same sync. A record whose status changed is rewritten without re-embedding. Search responses show the index build time.
Output and evidence	Result type, provenance chip, published and collected time, why it matched (keyword or meaning).
Failure modes	Permission leak through post-filtering; stale index after a rebuild; exact names lost to semantic neighbours.
Existing modules	`src/lib/assistant/lance.ts`, `embed.ts`, `api/collector/search` [S]
Evaluation	Recall@10, MRR and nDCG@10 per language; leak test (an officer query never returns other-department items).
Proposed acceptance	0 permission leaks in the leak test; metrics reported per language; switch embedding model only if it beats the current one beyond the confidence interval.
Effort, dependencies, risk	M. Low risk for fields; medium for an embedding switch (full re-index).
Decision	**Implement now** (fields, filters, exact lane, story collapse); **benchmark first** (bge-m3, reranker).

4.E Natural-language analytics and conversation

Field	Detail
Administrative question	"Which areas had the most serious incidents last week? Why there? Only police. Compare with the week before. Show the records."
Current implementation	[S] LLM router (falls back to keyword rules) → tools or a plan compiled to SQL → composer → verifier. Typed `ConversationContext` with `ActiveFilters` (period, zone, taluk, dept, cat, topic, sev, openOnly, overdue), result IDs and selected incident.

Field	Detail
Measured weakness / uncertainty	[V] The rules fallback loses refinements and misroutes brief examples (`AI-002`). Live LLM routing **not evaluated**. No golden conversation tests (notes say only intents are tested [D]).
Recommended approach	Make conversation state explicit and auditable. (1) Each turn produces a **state delta**: `set`, `keep`, `clear` per field (period, comparison period, geography with its level, department, category, severity, status, selected incident, result IDs). (2) The router outputs the delta, not a whole new scope; code applies it. (3) Explicit resets: "start over", "all departments", "whole district". (4) Show the active filters as chips above each answer, so the Collector sees what was kept. (5) Ambiguity: when two readings change the answer (e.g. "PWD" = state PWD or GCC works), ask once with chips. (6) Causal "why" questions return the drivers the data can show (which categories, areas or sources changed), plus an explicit evidence limit: "the data shows where the increase is, not its cause".
Why it fits	It builds on the existing typed context; it makes the LLM's job smaller and checkable.
Alternative and trade-offs	Let the LLM rewrite the full question each turn ("query rewriting"). Simpler, but filters drift silently. **Not recommended** as the source of truth.
Inputs and labels	15 scripted conversations × 4–6 turns, written with expected state after each turn; 30% Tamil or Tanglish.
Date interpretation	"Today", "yesterday" and "last 7 days" are resolved against the **current clock** in IST (the console's `asOf()` already does this [S]). Never silently substitute the dataset's newest date. When the newest data in scope is older than the feed's expected refresh, the answer says so ("news last collected 6 h ago; today's figures may be incomplete"). The dataset's date is used only when the user asks for it ("as of the last update"). Store explicit start and end times and show them ("Last 7 days: 1–7 Oct"). "This week" vs "last 7 days" is clarified when it changes the answer.
Failure modes	A new place mentioned without "only" (replace or add?); comparisons whose two periods use different filters.
Existing modules	`src/lib/assistant/context.ts`, `decide.ts`, `pipeline.ts`, `intent.ts` [S]
Evaluation	Share of turns whose resulting state matches the expected state exactly; share of answers whose SQL scope equals the state.
Proposed acceptance	Conversation state exactly correct on ≥ 90% of turns (T9); no answer may use a scope different from the chips it displays (automated check, 100%).
Effort, dependencies, risk	M. Depends on scoped facts (4.F). Medium regression risk in a 1,469-line pipeline; add golden tests first.
Decision	**Implement now**, behind golden tests.

4.F Numerical accuracy and evidence

Why a matching number is not enough. The current verifier [S] accepts a number if some fact has the same value at the stated precision, adjusted for measure words and direction. Example of a sentence that would pass incorrectly: "Zone 9 had 12 open complaints this week" when 12 is in fact Zone 5's count, or last week's count. The value exists in the facts but does not support *this* sentence's subject, scope or time.

Field	Detail
Administrative question	"Is every number I'm told the right count, of the right thing, for the right place and period?"
Current implementation	[S] `Fact {id, label, value, unit}`; `verifyNumbers` (value, rounding, measure words, direction); `{{fact_id}}` placeholders filled by code. [D] Briefings report "unverified numbers: 0".
Weakness	[S] Facts carry no scope: no filters, window, unit of counting, or denominator. Verification checks presence, not support.
Recommended approach	**Scoped facts and claim binding.** (1) Each fact records: measure (incidents, reports, complaints, deaths…), **counting unit** (event vs report vs complaint), filters (geography with level, department, category, severity, status, provenance), window (start and end in IST), denominator (for rates and shares), comparison basis, query ID and source row count. (2) The composer must cite a fact ID for every number (already supported by `{{fact_id}}`). The verifier then checks the sentence's named place, period and measure against the fact's scope, not just its value. (3) Percentages

Field	Detail
	carry numerator and denominator facts. (4) Rounding rules are fixed per unit. (5) Comparisons require both facts with identical filters except the period.
Why it fits	It extends the existing grounded-citation design. Code still computes; the model only arranges.
Alternative	A second LLM as a checker. Cheaper to build but unreliable and quota-hungry. **Not recommended** as the gate; possibly an extra signal.
Unit-of-count rules	Incidents = distinct incident IDs; reports = documents; complaints = citizen complaint records (one incident can have many). Independent corroboration = distinct publisher groups, not documents.
Failure modes	Synonyms for places (Tamil and English names), "last week" vs "previous 7 days", shares without a stated denominator.
Existing modules	`src/lib/assistant/verify.ts`, `facts.ts`, `types.ts`, `compile.ts` [S]
Evaluation	On 60 reviewed answers: number correctness (value equals SQL on the frozen build), scope correctness (sentence scope equals fact scope), citation support (a person agrees the cited records support the sentence).
Proposed acceptance	0 wrong-scope numbers and 0 unsupported numbers in the reviewed set; any failure blocks release.
Effort, dependencies, risk	M (3–5 days). Low–medium risk: more answers fall back to templates at first.
Decision	**Implement now.**

4.G Prioritisation and briefings

Three different things are currently blended. Separate them and show all three:

Axis	Question	Basis
Severity	How bad is the event itself?	Deaths, injuries, hazard, people affected (from evidence spans)
Urgency	How soon must someone act?	Ongoing vs over, deadline breach, growth, vulnerable people, weather
Confidence	How sure are we that it happened as described?	Official record, number of independent outlets, provenance, extraction method, location precision

Field	Detail
Administrative question	"What needs my attention first, and how sure are we?"
Current implementation	[S] Severity rules; priority = severity + hand-set weighted signals, with reasons; learning idle (0 decisions [V]). News-only incidents capped at High until confirmed [V].
Recommended approach	(1) Keep rules (explainable; no outcome labels exist). (2) Compute the three axes separately and rank by urgency within severity bands, displaying confidence. (3) Unconfirmed serious reports: severity can be Severe *as reported* with confidence "unconfirmed". They go in a separate lane, never mixed into confirmed counts. (4) Briefing structure: changes since the previous briefing (new, escalated, closed, stalled); serious-unconfirmed; department attention (overdue and stalled actions); hotspots from located records; each line with evidence and grade. (5) Corroboration counts independent publisher groups only.
Alternative	Learned ranking (learning-to-rank) from Collector decisions. **Defer**: needs outcome labels; the existing MAP refit becomes eligible at 30 decisions.
Failure modes	Confidence presented as truth; a widely syndicated rumour ranked as corroborated.
Existing modules	`district_intel/dintel/severity.py`, `priority.py`, `agents/briefing.py`; web `intel.ts`, `BriefingBook.tsx` [S]
Evaluation	Two reviewers rank 30 incidents blind; measure agreement with the system order (Kendall's tau) and with each other.
Proposed acceptance	Kendall's tau between system order and each reviewer ≥ 0.6, and within the reviewer–reviewer range (T12).

Field	Detail
Effort, risk	M; low risk.
Decision	**Implement now** (axes and lane); **defer** (learned ranking).

4.H Anomalies and early warnings

Field	Detail
Administrative question	"Is something unusual happening, beyond normal variation, and is it real or just a change in reporting?"
Current implementation	[S] Per category and zone: Poisson expected count over 28 days with weekday and rain factors; one-day spike tail test and Poisson CUSUM slow rise (p-value by simulation); Benjamini–Hochberg at 5% false discovery rate across all checks. [V] 3,824 checks → 5 spikes, 1 slow rise (4 Oct build).
Assessment	The statistics are appropriate and conservative. The **data** is the problem: series are dominated by synthetic records, and rain factors are learned from them.
Recommended approach	(1) Retain the method. (2) Run it on real-only series where they exist (news incident counts by category). Present synthetic-series alerts as "demo data". (3) Add a **coverage guard**: before raising an alert, check that source volume (articles fetched, feeds healthy) did not change by a similar factor; otherwise label "possible collection change". (4) Use an exposure denominator for news (incident articles per 1,000 Chennai articles) so a busier news day isn't read as more incidents. (5) Small counts: keep the minimum count (3) and show the expected value next to the observed. (6) Reporting delay: exclude the last few hours of the current day, or mark it provisional.
Early warning	**Defer** forecasting of complaints until real complaint history exists. Real rainfall and river-level threshold alerts from IMD and CFM can be shown now, as observations, not predictions.
Alternative	Prophet-style seasonal models or Bayesian structural time series: need longer real history. **Defer.**
Evaluation	Weekly review of all alerts by a person (true signal / collection artefact / noise); alert precision over 8 weeks.
Proposed acceptance	Every alert shows observed, expected, q-value and coverage check; precision reported after 8 weeks of real data.
Decision	**Implement now** (coverage guard, real-only series); **defer** (forecasting).

4.I Intervention analysis

Field	Detail
Administrative question	"Did desilting this drain actually reduce flooding complaints?"
Why before/after is not enough	Rain, season, festivals, reporting campaigns and regression to the mean all change counts with no action taken.
Recommended design	Difference-in-differences: compare the change in the treated area with the change in matched control wards (similar baseline rate, low-lying index, zone) over the same windows. Require a minimum number of events before and after. Report an interval, not a single number. If the interval includes zero, or data is insufficient, the outcome is "**insufficient evidence**".
Confounders to record	Rain days in each window, other actions in the same ward, changes in source coverage.
Data needed	Real complaint data with ward geography; action records with dates and wards.
Decision	**Defer** until real departmental feeds exist. Document the method now so it isn't improvised later.

5. AI model selection

5.1 What kind of method each task needs

Task	Deterministic code	Statistical model	Classifier	Embeddings	Reranker	LLM	Human review
Counts, KPIs, comparisons	**Yes (only)**	—	—	—	—	No	Spot checks
News district relevance, report type, category	Rules for exclusions	—	**Yes**	Optional features	—	Grey band and teacher	Gold labels
Entities, deaths, status, event time	Validators	—	—	—	—	**Yes**, with evidence spans	Field audit
Event place	Gazetteer, point-in-polygon	—	Role model (LR)	—	—	Optional	Gold labels
Same-event grouping	Exact-copy rules	Constrained agglomeration	Pair LR	**bge-m3**	**bge-reranker-v2-m3**	Grey band	Pair labels
Cross-source linking	Blocking	—	Pair LR	e5 similarity	—	Review band	Pair labels
Search	Exact lane	Fusion (RRF)	—	**Yes**	Optional	—	Query judgements
Assistant routing and planning	Compiler, validators	—	Rules fallback	—	—	**Yes**	Golden tests
Answer writing	Templates fallback	—	—	—	—	**Yes**	Answer review
Anomalies	—	**Yes**	—	—	—	No	Alert review
Priority	**Rules**	Later: learned weights	—	—	—	No	Ranking review
Translation of headlines	—	—	—	—	—	**Yes**	Sample check

5.2 Model comparison (candidates are hypotheses until benchmarked)

Task	Current model	Candidate	Why consider it	Language suitability	Hosting / privacy	Resource needs	Availability verified?	Benchmark	Decision
News classification (local)	TF-IDF + LR [S]; SetFit on multilingual-e5-base [S]	1) Same TF-IDF + LR retrained on reviewed gold. 2) Fine-tuned `google/muril-base-cased`	1) Fixes the label problem cheaply. 2) Indian-language pretraining incl. Tamil and transliterated text [E]	MuRIL: 17 Indian languages incl. Tamil; transliterated data [E]	Local; Apache-2.0 [E]	TF-IDF: seconds on CPU. MuRIL: BERT-base fine-tune, slow on CPU [H]	Model card fetched 8 Oct 2026	Same gold test set, per language	Retrain TF-IDF first; benchmark MuRIL
News classification (teacher, grey band)	Groq `openai/gpt-oss-120b` → `gpt-oss-20b` [S config]	Gemini flash models already configured [S]	Spread quota; compare structured-output validity	Tamil quality unmeasured [H]	Hosted; article text leaves the PC (public news, low sensitivity)	Free tier: Groq page lists 30 RPM, 8K TPM, 200K TPD base limits per model [E]	Groq limits page fetched; org limits may differ [E]	Validity and accuracy on 200 gold items	Keep as teacher only; measure
Extraction (deaths, status, people)	Groq gpt-oss (details) [S]	Same models with strict schema plus mandatory evidence span	Make outputs checkable	As above	Hosted	Same	—	Field accuracy on 150 full-text articles	Change prompt and schema, not model
Same-event retrieval	`BAAI/bge-m3` [S]	Keep	MIT, 100+ languages, 8,192 tokens, dense/sparse/multi-vector [E]	Multilingual incl. Tamil (listed as >100 working languages) [E]	Local	1024-d; CPU ~4.5 texts/s [D]	Card fetched	Pair recall at candidate stage	Retain

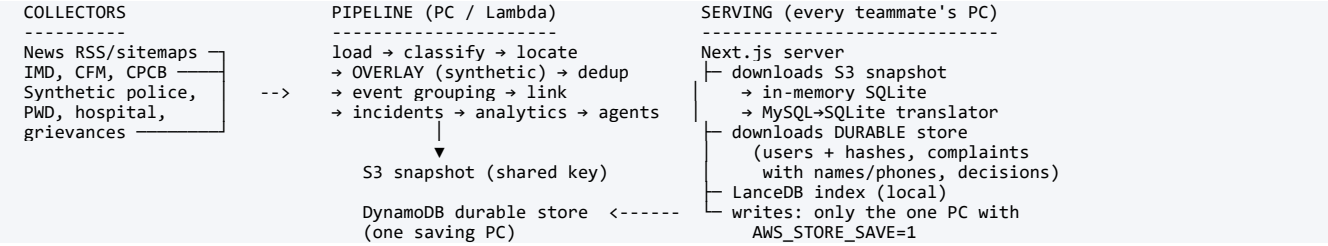
Task	Current model	Candidate	Why consider it	Language suitability	Hosting / privacy	Resource needs	Availability verified?	Benchmark	Decision
Same-event pair scoring	`BAAI/bge-reranker-v2-m3` + LR [S]	Keep; retrain LR on reviewed pairs	Apache-2.0, 0.6B, multilingual, based on bge-m3 [E]	As bge-m3 [E]	Local	CPU ~6 pairs/s [D]	Card fetched	Reviewed pairs, hard negatives	Retain
Linking features	`intfloat/multilingual-e5-small` [S]	bge-m3 vectors (already computed for news)	One model fewer	—	Local	Larger vectors	—	Real link pairs	Evaluate after labels exist
Web search embeddings	`Xenova/multilingual-e5-base` (768-d, ONNX) [S]	`BAAI/bge-m3`	Consistency with pipeline; longer input [E]	e5-base: 94–100 languages, 512 tokens, needs query/passage prefixes [E]	Local	Re-index; heavier CPU	Cards fetched; ONNX port of bge-m3 for transformers.js **not verified**	60 judged queries per language	Benchmark before switching
Search reranking	RRF only [S]	Cross-encoder on top 20	LanceDB supports cross-encoder rerankers [E]	bge-reranker multilingual [E]	Local	Added latency per query [H]	Docs searched 8 Oct 2026	Same 60 queries, latency recorded	Only if retrieval shows a gap
Assistant router/planner/composer	Groq gpt-oss-20b/120b, qwen3.8-27b; Gemini flash family [S defaults]	Keep pool; measure per-model schema validity and accuracy	Free tier, fallback already built	Tamil/Tanglish answer quality unmeasured [H]	Hosted; questions and facts leave the server (no PII by design [S])	Rate limits per model [E]	Partly (Groq page)	100-question golden set, frozen data	Retain; route by measured validity
Speech to text	Browser speech; `whisper-large-v3-turbo` configured option [S]	—	Voice is optional in the brief	Tamil STT quality unmeasured	Hosted/browser	—	Not verified	30 spoken commands	Low priority

Notes on the comparison:

- **No candidate is asserted to be better.** Each row ends in a controlled comparison on District IQ's own reviewed data, with the current model as the baseline.
- **Embedding switches** mean re-encoding and re-indexing every record. Keep the old index until the new one wins.
- **Free tiers** are shared by the news pipeline and the assistant (notes [D]). A per-task quota budget should be set so news labelling cannot starve the assistant, or the reverse.

6. Architecture, security and data reliability

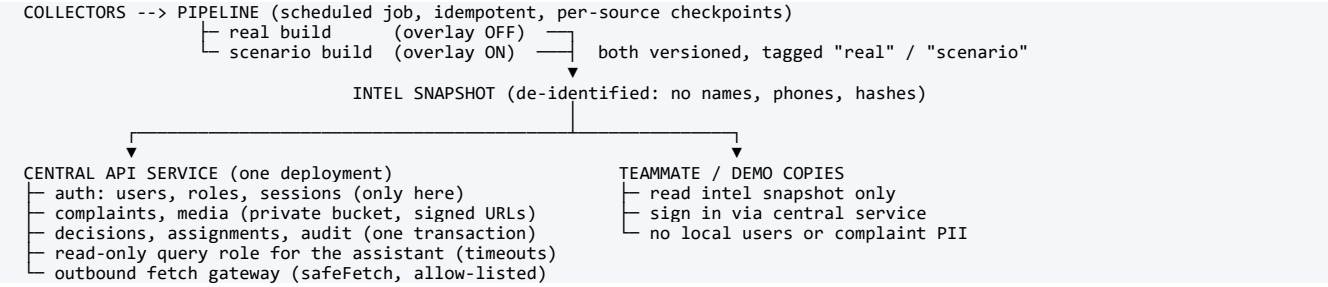
6.1 Current flow (as found)



Problems visible in this flow:

- Personal data on every machine (SEC-001).
- A single writer (data loss when that PC is off).
- Two SQL dialects (CODE-001).
- Reload pauses (NEW-11, a document claim).
- Synthetic overlay mixed into the same build as real data (DATA-001).

6.2 Proposed target flow



Effect on login and workflows:

- Teammate copies stop holding a `users` table. Sign-in goes to the central service (or a local development account file that holds no real users).
- Officers and the Collector use the central deployment for real work.
- Citizen filing writes only to the central service.
- This adds one hosted service, and is the only way to stop PII copies, rather than just hiding them behind different keys.

6.3 Specific recommendations

Concern	Recommendation	Verification
Central auth and roles	One auth service; per-role permissions enforced on the server for every route and inside search pre-filters; department from the database (already done for officers [S])	Route-by-route permission test; officer search leak test
Citizen privacy, password hashes	Never export `users` or complaint PII outside the central service; de-identified complaint views (ward, category, time, status) for intelligence	Column scan of every distributed artefact
Credentials	Per-person and per-service keys; rotation; separate read and write scopes	Old shared key returns 401
Uploads and media	Authenticated upload, magic-byte check, size and rate limits, private storage, signed time-limited URLs	Anonymous upload 401; direct URL without signature 403
Outbound fetch	One `safeFetch` for all link fetching: scheme check, DNS resolution with private-range and IPv6-mapped checks on every redirect hop, size and time limits, robots compliance	Unit tests with private, mapped and redirecting URLs

Concern	Recommendation	Verification
Durable writes, concurrency	Single central database with transactions; optimistic concurrency (row version) on decisions and assignments	Two-user conflicting update test
Stable identity	Keep `stable_ids`; add an alias table for merges and splits; never reuse an ID	Rebuild unchanged data 3× → identical IDs
Incremental ingestion	Per-source checkpoint (last item ID, ETag or time); idempotent upsert by source record ID plus content hash	Re-running a job twice changes nothing
Partial failures	Each source job independent; a failed source keeps its last good data, marked stale	Simulated feed failure shows a stale banner, not empty data
Provenance and snapshots	Store the raw fetched payload hash and fetch time per item (`snapshot_sha256` exists on events [V]); keep raw copies for 90 days	Any displayed item traces to a raw snapshot
Timestamps	Four fields: `event_time` (with precision), `published_at`, `collected_at`, `updated_at`; all stored in UTC, shown in IST	Schema check; display check
Audit durability	Decision and audit row in one transaction; failure aborts the decision	Forced audit failure test
Read-only queries	Separate read-only DB role or read-only SQLite handle; statement timeout; cancellation when the user asks a new question	Write attempt fails; long query cancelled
MySQL/SQLite	Choose one serving engine. Option 1: SQLite file served read-only (simple, matches pipeline output). Option 2: MySQL/Postgres for the central service plus SQLite snapshot for intel. Avoid translating dialects at run time.	Parity test until migration completes
Caching	Cache intel reads keyed by build ID; never cache decision-dependent counts beyond the decision version; invalidate on any decision	Decision → tile updates immediately
Store reload pauses	Load a new build in a worker or separate process and swap a handle; never block request threads	Measure p95 response time during a reload

7. Real-data evaluation plan

Principles:

- **Gold labels** are made by two people working independently. A third person resolves disagreements, and agreement (Cohen's kappa) is reported.
- **AI drafts** may pre-fill a label but are hidden from the second labeller.
- **Freeze** the dataset (build ID) used for each evaluation.
- **Split** data into training/tuning items from **earlier dates**, with the test set from a **later week**, to test generalisation and prevent leakage. Duplicate stories must not straddle the split.
- **Never invent expected numbers.** Compute them from the frozen build.
- **Sample sizes:** a proportion near 0.8 with a ±5 percentage-point margin at 95% confidence needs about 250 items, and ±7 points needs about 125 ($n \approx 1.96^2 \times p(1-p) / e^2$).
- **The margin applies only to the group that has that many items.** A 300-item test split across three languages gives about 100 per language, and each language result is then only about ±8 points. Categories are smaller still. Every subgroup result is reported with its own n and interval. Any subgroup under 100 items is marked "indicative", and no pass/fail decision rests on it alone.
- **Precision and recall have different denominators.** Precision needs enough predicted positives and recall enough true positives. Stratify the sample so both reach about 100 where a threshold depends on them.

Area	Sampling	Size and rationale	Labelling	Metrics	Baseline	Proposed target
District relevance and incident filter	Stratified by language, publisher tier, classifier confidence band; later week for test	900 (600 tune / 300 test with 100 per language); overall ±5 pts, per language ±8–10 pts	Two labellers, adjudication; written guide (existing README rules)	Precision, recall, F1 with CI; per language	Current cascade on same test	T1–T2 overall; per-language results reported, marked indicative when n < 100
Category and report type	Same items	Same; rare categories (< 20 test items) pooled as "other" for scoring	Same	Macro-F1; confusion matrix	Current	T3: macro-F1 ≥ 0.70
Extraction fields (deaths, injured, status, event time)	Full-text incident articles only	150	Spans marked by labellers	Field accuracy; unsupported-field rate	Current LLM details	Unsupported fields = 0 (span rule)
Event place	Reports naming ≥1 district place; include hospital and dateline cases	300 (200 tune / 100 later test)	Two labellers	Accuracy; wrong-place rate; coverage	Current role model	T4: wrong-place ≤ 10%
Same-event pairs	Candidate pairs by cosine band; 25% Tamil–English; 30% hard negatives	400	Two labellers; unsure excluded and counted	Pair precision/recall; false-merge rate on hard negatives	Current thresholds	T5: false merges ≤ 5%
Story clusters	Whole stories, random and largest	60 stories	One labeller + check	B-cubed precision/recall	Current	T6: B-cubed precision ≥ 0.85
Cross-source links	Existing 141 sampled pairs + more across bands	150–300	Two labellers	Precision/recall by band, band-weighted	Current LR (trained on planted pairs)	T7: precision ≥ 0.80 in the auto-link band; recall reported
Retrieval	Real queries written by people, not from index text	60 (20 per language)	Relevance judgements per result	Recall@10, MRR, nDCG@10; leak test	Current e5-base + BM25 + RRF	T8: Recall@10 ≥ 0.80 overall; a new model must beat the current one beyond the interval
Analytics answers	Questions from the brief plus Collector-style questions	60	Expected values computed by SQL on frozen build	Value correctness; scope correctness	Current assistant	0 wrong-scope numbers
Follow-up conversations	Scripted chains	20 × 4–6 turns (about 100 turns)	Expected state per turn	State exact-match rate	Current	T9: ≥ 90% of turns

Area	Sampling	Size and rationale	Labelling	Metrics	Baseline	Proposed target
Citation support	Answers with citations	60	Reviewer judges support	Supported-citation rate	Current	100% supported in reviewed set
Safe refusals	Unsafe, injection, and near-miss safe questions	30 unsafe + 50 near-miss	Labelled	Unsafe-allowed rate; false-refusal rate	Audit: 9/10 refused; 2 false refusals	0 unsafe allowed; 0 false refusals
Tamil and Tanglish	Included in each set above	≥30% of items	Native speakers	Per-language metrics	—	Gap to English reported
Failure handling	Simulated: model unavailable, feed stale, store reload	10 scenarios	Expected behaviour written	Pass/fail	—	All pass

7.1 Provisional thresholds (agree before implementation is approved)

These numbers are **targets proposed for agreement**, not measured results. They should be signed off before work is approved, then confirmed or tightened once the first baseline is measured. A target that turns out to be unreachable is renegotiated in writing, not silently dropped.

ID	Metric	Provisional threshold	Applies to
T1	Chennai-incident precision (news filter)	≥ 0.85	Overall gold test set
T2	Chennai-incident recall	≥ 0.75	Overall gold test set
T3	Category macro-F1	≥ 0.70	Categories with ≥ 20 test items
T4	Wrong event place	≤ 10%	Place test set
T5	False merges on hard negatives	≤ 5%	Pair test set
T6	Story cluster B-cubed precision	≥ 0.85	60 reviewed stories
T7	Cross-source link precision in auto-link band	≥ 0.80	Real labelled pairs
T8	Search Recall@10	≥ 0.80	60 judged queries
T9	Conversation state exactly correct	≥ 90% of turns	About 100 scripted turns
T10	Wrong-scope or unsupported numbers	0	60 reviewed answers
T11	Unsafe requests allowed / false refusals	0 / 0	30 unsafe + 50 near-miss
T12	Priority order agreement (Kendall's tau)	≥ 0.6	30 incidents, 2 reviewers

Regression checks:

- Every evaluation set becomes a test that runs on each release.
- A metric drop beyond its confidence interval blocks the release.
- The pipeline's `evaluation.md` reports gold-set metrics separately from synthetic-truth metrics, with sample sizes and labeller agreement.

Labelling effort (estimate): about 2,300 items at 1–2 minutes each, done twice, plus adjudication. That is roughly 9–14 person-days, which can be split among team members.

8. Implementation roadmap

Assumptions:

- Estimates are developer-days for someone who knows the codebase.
- Labelling is separate person-days.
- Ranges include normal debugging but not waiting on outside parties (boundary files, AWS access).
- A second developer does not halve the time: shared modules (`intel.ts`, `pipeline.ts`) and review limit parallel work. Expect roughly 60–70% of the one-developer elapsed time [H].

#	Change	Affected modules (verified by reading)	Depends on	Effort (dev-days)	Data prerequisite	Regression risk	Rollback	Validation gate	Collector benefit	Demo value
A1	Provenance end to end (screens, exports, index, facts); real vs scenario builds	`intel.ts`, `Overview.tsx`, `Detail.tsx`, `reportPdf.ts`, `lance.ts`, `world.py`	—	3–5	—	Low	Feature flag	Real-only: Severe = 0 on current data; every view shows a chip	Trust	Very high
A2	Central auth and personal data; stop distributing PII and hashes	`lib/aws/store.ts`, AWS API, `auth.ts`	AWS access	5–8	—	High	Keep old path behind flag until verified	No PII in any distributed file	Privacy	High
A3	Private authenticated uploads and media	`api/complaints/media`, file-complaint page	A2 (auth)	1–2	—	Medium	Revert route	Anonymous 401; signed URLs	Safety	Medium
A4	Precision-aware geography (radius, no coarse pins, unlocated bucket, roles shown)	`intel.ts`, `SatMap.tsx`, `placerole.py` outputs	—	1–2	—	Low	Flag	0 coarse pins; totals reconcile	Map honesty	High
A5	Effective status everywhere; read-only GETs; routing job with audit	`intel.ts` (`kpiBlock`, `routeNewsComplaints`), `sources.ts audit()`	—	2–3	—	Medium	Revert	Counts match across views; no writes on GET	Consistent numbers	Medium
A6	Guard false refusals + tests	`guard.ts`, guard tests	—	1	—	Medium	Revert	0 false refusals; unsafe still refused	Usability	High
A7	One safeFetch	`onboard.ts`, `sources.ts`, `fulltext.ts`, `studio/fetchurl.ts`	—	1–2	—	Low	Revert	SSRF tests pass	Security	Low
A8	Fix failing tests; commit grouping code; CI for pytest	`eventmatch.py`, `loaders/news.py`, tests	—	1–2	—	Low	Git	All suites green in CI	Reliability	Low
A9	Durable audit; secure defaults; persisted limits	`sources.ts`, `jwt.ts`, `db.ts`, `rate-limit`, `gateway.ts`	—	1–2	—	Low	Revert	Forced-failure tests	Accountability	Medium
B1	Gold labelling and evaluation harness	`newseval.py`, `linking.py` label tools; new eval runner	—	3–5	9–14 person-days labelling	None	—	Kappa reported; harness in CI	Proven accuracy	Very high
B2	Retrain news cascade on gold; shrink review backlog	`newslocal.py`, `newsllm.py`	B1	3–5	Gold set	Medium	Keep old model file	Gold-test metrics ≥ baseline; review share down	More real incidents found	High
B3	Grouping: exact-copy rule, backlog, thresholds, update links, split/merge review with aliases	`eventmatch.py`, `loaders/news.py`, `linking.py`	A8, B1	3–5	Pair labels	Medium	Alias table preserves IDs	False-merge limit; IDs stable	Correct counts	High

#	Change	Affected modules (verified by reading)	Depends on	Effort (dev-days)	Data prerequisite	Regression risk	Rollback	Validation gate	Collector benefit	Demo value
B4	Cross-source linking on real labels; "no matching record found" semantics	`linking.py`	B1	2–3	Link labels	Medium	Old bands	Real-pair metrics reported	Honest gaps	Medium
B5	Scoped facts and claim verification	`verify.ts`, `facts.ts`, `types.ts`, `compile.ts`	—	3–5	—	Medium	Old verifier as fallback	0 wrong-scope numbers in reviewed set	Correct numbers	High
B6	Conversation state deltas, chips, resets, golden conversations	`context.ts`, `decide.ts`, `pipeline.ts`	B5	3–5	Scripted chains	Medium	Flag	State match rate measured	Coherent follow-ups	High
B7	Read-only query handle and cancellation	`compile.ts`, `store.ts`	—	2–3	—	Medium	Revert	Write fails; cancel works	Stability	Low
B8	Extraction with evidence spans; event time separate from publication	`newsllm.py`, schema	B1	3–4	150 reviewed articles	Low	Old fields kept	0 unsupported fields	Reliable details	Medium
C1	Serious-unconfirmed lane; three axes; independent-outlet corroboration	`severity.py`, `priority.py`, `intel.ts`	A1, B3	2–3	—	Low	Flag	Every real fatal report appears; syndication counted once	Real events surface	Very high
C2	Changes since last briefing	snapshot table, `intel.ts`, briefing	A5	3–4	—	Low	Hide strip	Matches manual diff on 3 days	Fast morning scan	High
C3	Per-item freshness and stale banner	`intel.ts`, shell	—	1–2	—	Low	Hide	Banner within one cycle	Trust	Medium
C4	Assignment deadlines, escalation, closure evidence	officer and collector ops tables	A2, A5	3–5	—	Medium	Flag	Loop completes with audit	Accountability	High
C5	"No matching record found" board	`intel.ts`	B4	2–3	—	Low	Hide	Count = SQL; recall shown	Gaps surfaced	High
C6	Unified search with permission and provenance pre-filters	`lance.ts`, search API	A1	3–5	60 judged queries	Medium	Old index kept	0 leaks; metrics reported	One search box	High
D1	Non-blocking store reload	`store.ts`	—	2–4	—	Medium	Revert	p95 latency during reload measured	Responsiveness	Low
D2	Build-keyed caching	`intel.ts`	A5	1–2	—	Low	Disable cache	No stale decision counts	Speed	Low
D3	Single serving engine or parity CI	`dialect.ts`, `store.ts`	A2	2–3 (parity CI)	—	Medium	—	Parity green	Reliability	Low
E1	Evidence-graded briefing	briefing, `BriefingBook.tsx`	A1, C1, B1	3–5	—	Low	Flag	Every line graded and linked	Faster judgement	High
E2	Tamil voice brief	`speech.ts`	E1	1–2	—	Low	Hide	Spoken numbers match screen	Accessibility	Medium

Totals (sum of the rows above):

Phase	Rows	Dev-days
A	A1–A9	16–27
B	B1–B8	22–35

Phase	Rows	Dev-days
C	C1–C6	14–22
D	D1–D3	5–9
E	E1–E2	4–7
All		**61–100 dev-days**, plus 9–14 person-days of labelling

For one developer that is about 12–20 working weeks; for two, roughly 7–14 weeks elapsed. This two-developer figure is a guess, not a measurement [H]: shared modules and review limit how much work can run in parallel.

The first five changes: A1 (provenance), A2 (central personal data and auth), B1 (gold labels and harness), A4 (precision-aware geography), C1 (serious-unconfirmed lane). C1 depends on B3 (grouping, so syndicated copies are counted once), which depends on A8 (failing tests). Including those prerequisites: A1 3–5 + A2 5–8 + B1 3–5 + A4 1–2 + A8 1–2 + B3 3–5 + C1 2–3 = **18–30 dev-days**, plus labelling. A faster option ships a first version of C1 without independent-outlet grouping, labelled "corroboration not yet de-duplicated" (saves B3: 15–25 dev-days).

Minimal credible challenge-ready scope: A1, A3, A4, A5, A6, A8, B1 (a reduced set: 300 incident items, 200 pairs, 150 places), B3, B5, C1, C3, plus A2 at minimum to the point that no distributed copy holds password hashes or citizen PII. Sum: A1 3–5 + A2 5–8 + A3 1–2 + A4 1–2 + A5 2–3 + A6 1 + A8 1–2 + B1 3–5 + B3 3–5 + B5 3–5 + C1 2–3 + C3 1–2 = **26–43 dev-days**, plus about 5 person-days of labelling.

Fuller product-excellence scope: all rows above.

Wait for real departmental feeds:

- Within-source dedup re-evaluation.
- Rain-to-complaint forecasting.
- Intervention analysis.
- Learned priority weights.
- Department performance league tables.
- Any claim that "departments did not record X".

9. What to avoid

Tempting choice	Why it is unsuitable here	Do instead
A larger LLM for classification or answers	The bottlenecks are unreviewed labels, quota and scope errors, not model size; a larger model also uses more of the free quota	Retrain local models on gold; measure LLM validity per task
Replacing bge-m3 / reranker / constrained grouping for novelty	Appropriate design for Tamil–English events; replacing it discards working code without evidence	Retune thresholds on reviewed pairs
Merging on headline similarity alone	Same-day same-type events have near-identical headlines	Exact-copy rule only; everything else through place, time and pair model
Showing model confidence as truth	Confidence is calibrated (at best) on its training data, not proof the event happened	Show evidence: sources, independence, official record
Predictions validated on synthetic data	They reflect the generator's assumptions	Defer, or label "demo model"
"Missing from departmental records" claims	Departmental data is synthetic; link recall on real data unknown	"No matching record found", with measured recall
Counting syndicated copies as corroboration	One agency story republished is one source	Count independent publisher groups
Kubernetes, microservices, streaming platforms	No measured load problem; adds operational risk for a small team	One central service and scheduled jobs
Security controls that break workflows (e.g. locking officers out of photos they need)	Users route around controls	Design central auth with the officer, Collector and citizen flows tested end to end
Promising accuracy or "10/10"	No reviewed real-data measurement exists yet	Publish measured metrics with sample sizes and intervals
Treating passing unit tests as proof of accurate answers	Tests check code paths, not real-world correctness	Gold-set evaluations as release gates

10. Final decision table

Problem	Recommended algorithm / model / architecture	Retain or change?	Reason	Evidence still needed	Priority
Synthetic data (permitted) is not labelled	Provenance fields end to end; separate real and scenario builds	Change (extend)	Use is allowed; unlabelled display misleads (DATA-001)	None to start	P0
Personal data on every copy	Central auth and data service; de-identified intel snapshot	Change	Key splitting alone still copies PII (SEC-001)	Hosting choice	P0
Unknown places shown as exact	Uncertainty radius, geo level, no coarse pins	Extend	GIS-001	—	P0
Accuracy unproven on real data	Human-reviewed gold sets and evaluation harness	Change (process)	Labels are AI drafts or empty (NEW-1, NEW-2)	Labelling time	P0
Real serious events hidden	Three axes; serious-unconfirmed lane; independent corroboration	Extend	SEV-001	—	P0
Numbers right value, wrong scope	Scoped facts + claim binding	Extend verifier	NEW-8	Reviewed answer set	P1
News backlog unlabelled (70%)	Retrain TF-IDF+LR cascade on gold; LLM grey band only; benchmark MuRIL	Retain method, retrain	NEW-4	Gold labels; MuRIL benchmark	P1
Same-event grouping	bge-m3 → bge-reranker-v2-m3 → LR → LLM → constrained agglomeration	Retain; retune	Sound design; unproven thresholds	Reviewed pairs	P1
Cross-source linking	Blocking + LR, retrained on real pairs	Retrain	Trained on planted pairs	150–300 real labels	P1
Follow-up drift	Typed state deltas, visible chips, explicit resets	Extend	AI-002	Golden conversations	P1
Guard false refusals	Command-form regex + tests	Fix	AI-001	—	P0
Search gaps	Index fields, pre-filters, exact lane, story collapse; benchmark bge-m3	Extend; evaluate	NEW-6, NEW-7	60 judged queries	P1
Taluk wrong source	Official polygons	Replace	GIS-002	Boundary file	P2
Anomaly false signals	Poisson + CUSUM + BH, plus coverage guard, real-only series	Retain; extend	Method sound; data synthetic	8 weeks of reviewed alerts	P1
Forecasting, intervention effects	DiD with controls; forecasting deferred	Defer	No real outcome data	Real feeds	P2
Two SQL dialects, reload stalls	One serving engine; non-blocking reload	Change	CODE-001, NEW-11	Latency measurement	P1
GET writes, audit gaps	Jobs with transactional audit	Fix	BE-001, OPS-001	—	P0

10.1 If we implement only five improvements

1. **Provenance everywhere, with a real-only mode** (`DATA-001`). Synthetic data is allowed; labelling it shows the judges the platform is transparent, and makes every later demo credible.
2. **Central handling of personal data and authentication** (`SEC-001`, with `SEC-002`). It removes the most serious real-world risk. Splitting keys is not enough; the data must stop being copied.
3. **Human-reviewed gold sets and an evaluation harness** (NEW-1, NEW-2, `LINK-001`). Every intelligence claim (relevance, place, grouping, linking, answers) currently rests on AI drafts or synthetic truth. This is the foundation for all algorithm work and the strongest answer to judges' accuracy questions.
4. **Precision-aware geography** (`GIS-001`). It removes the most visible false signal (the centroid "hotspot"), and makes area totals honest.
5. **The serious-but-unconfirmed lane with three separate axes and independent corroboration** (`SEV-001`). It brings real tragedies to the Collector's attention the same day without presenting them as confirmed. That is the most direct improvement in operational usefulness.

These five, with the grouping work (B3) and test fixes (A8) the lane depends on, need roughly 18–30 developer-days plus 9–14 person-days of labelling (a reduced first labelling round of about 5 person-days is possible). Algorithm retuning (B2–B4) should follow immediately, because the gold sets make it measurable.

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All web sources were accessed on 8 October 2026. Sources marked "not fetched" are standard references cited from knowledge; their availability was not re-checked.

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Glossary

Term	Meaning
B-cubed	A clustering score that checks, for each item, how many of its cluster-mates are correct (precision) and found (recall)
Benjamini–Hochberg (BH)	A way to run many statistical tests while keeping the share of false alarms among raised alerts near a chosen rate
Blocking	Comparing only plausible pairs (same category, nearby, close in time) instead of every pair
Calibration	Making a model's confidence match how often it is right (0.8 confidence → right about 80% of the time)
Constrained agglomeration	Building groups by joining the closest items, but refusing joins that would connect clearly different items
Cross-encoder / reranker	A model that reads two texts together and scores how well they match; slower but more accurate than comparing separate vectors
CUSUM	A running-sum test that detects a sustained rise rather than a one-day spike
Difference-in-differences	Comparing the change in a treated area with the change in similar untreated areas over the same time
Embedding	A list of numbers representing a text's meaning, so similar meanings are close together
Evidence span	The exact words in a source that support an extracted value
Gold / silver labels	Gold: human-reviewed correct answers. Silver: automatic or AI-drafted labels used for training, not for judging accuracy
Hard negative	A pair that looks alike but is truly different (two accidents on the same road), used to test false merges
Hybrid search	Combining keyword search and meaning-based search
Kappa (Cohen's)	Agreement between two labellers beyond chance
Logistic regression (LR)	A simple, explainable model that turns weighted features into a probability
MRR / nDCG / Recall@k	Search quality measures: how high the first right answer ranks; how good the whole ranking is; whether the right items appear in the top k
PII	Personally identifiable information: names, phone numbers, addresses, ID numbers
Point-in-polygon	Deciding which ward boundary contains a map point
Provenance	Where a record came from, and whether it is real, synthetic or derived
RRF (reciprocal rank fusion)	Merging ranked lists by position rather than raw scores
Schema-constrained output	Forcing a model to answer in a fixed structure (fields and types) that code can validate
SetFit	Fine-tuning a sentence-embedding model on few labelled examples, then adding a small classifier
SSRF	Tricking a server into fetching private or internal addresses
Synthetic / scenario data	Generated records standing in for unavailable departmental systems
TF-IDF	Word and character-pattern counts weighted by how distinctive they are; a fast, explainable text feature